AN-102 Thermal Design Guidelines

Halcyon Micro — fictional company, synthetic document

Junction temperature

The junction temperature is TJ = TA + PD × θJA, where PD is the power dissipated in the device and θJA is the junction-to-ambient thermal resistance of the package on a standard JEDEC board.

Linear regulator dissipation

For a linear regulator, PD = (VIN – VOUT) × IOUT plus the quiescent term VIN × IQ, which is negligible at load.

Worked example: HM3300

VIN = 5.0 V, VOUT = 3.3 V, IOUT = 400 mA gives PD = (5.0 – 3.3) × 0.4 = 0.68 W. In the SOT-23-5 package (θJA = 220 °C/W) the temperature rise is 0.68 × 220 = 149.6 °C, which exceeds the 125 °C limit even at TA = 25 °C. In the DFN-6 package (θJA = 65 °C/W) the rise is 44.2 °C, so TJ = 69.2 °C at TA = 25 °C and the design is within limits.

Package thermal resistance summary

Part	Package	θJA (°C/W)
HM3300	SOT-23-5	220
HM3300	DFN-6 (2×2 mm)	65
HM5V-PD	QFN-24	38
HT410	SOT-563	190
HMD12	TSSOP-20 (EP)	32
HIMU6	LGA-14	90
HF16	WSON-8	45
HF16	SOIC-8	125
HAC24	QFN-32	34

Layout recommendations

Use a solid copper pour under exposed pads, add thermal vias (0.3 mm drill, 1.0 mm pitch) beneath QFN and TSSOP exposed pads, and keep heat-generating devices away from the HT410 temperature sensor.